

Eval-50 gap report

Sessions analysed: 50

Top-line metrics

- Average reply length: 31.5 words
- Replies over 120 words: 0
- Replies under 8 words: 0
- Avg memory-reference rate (returning-user sessions): 35%
- Opener-repetition pairs (≥ 0.65 similarity, same user, different sessions): 1
- Total stale-phrase hits across all sessions: 18
- Sessions with a clean wrap-up signal: 27

Ranked gaps (severity x frequency)

1. `opener\_repetition` · severity 5 · freq 2.0% · score 250

Layer: guard

1 pairs of openers across same-user sessions had ≥ 0.65 structural similarity.

Examples:

- u01_priyansh sessions 5 <-> 8 (sim 0.68): Hi Priyansh, I heard you're working on something exciting with cricket gloves - how are the trial plans coming along? <-> Hi Priyansh, I heard about the cricket gloves project you're working on - sounds exciting! How are the trials coming alo

Suggested fix: Add a deterministic first-turn similarity check that strips opener if normalized form matches one of the last 6 stored opening_phrases by SequenceMatcher ≥ 0.7 .

2. `greeting\_on\_turn\_2plus` · severity 4 · freq 56.0% · score 224

Layer: prompt

Maya started a turn-2+ reply with a greeting ('Hi <name>,...') in 131 replies across 28 sessions.

Examples:

- session 1 turn 2: Hi Priyansh, that sounds fun! I love cricket and Bollywood - which one is your favorite, Pathaan or Tiger 3?
- session 2 turn 2: Hi Priyansh, that sounds exciting! Trying new positions is a great way to improve. Yuvraj Singh is a legend - you're on
- session 4 turn 3: Hi Priyansh, I'm glad you like Virat's movies - "entertaning" is a nice word! I watched Pathaan again last week; it's

Suggested fix: Strengthen Rule 1 turn-2+ no-greeting clause OR add a server-side strip in pg_apply_output_guard.

3. `multiple\_questions\_per\_reply` · severity 3 · freq 56.0% · score 168

Layer: prompt

Maya included 2+ questions in 53 replies across 28 sessions (Rule 9 violation).

Examples:

- session 5 turn 8: You explained that clearly! Good use of "very good." I agree - being a strong player helps! Me? I'd probably forget all the strategies. Do you enjoy strategizin
- session 6 turn 3: Hi Priyansh, I see you are thinking hard about the cricket gloves - simplifying is a smart idea. What about a Bollywood song to relax? I just listened to Ek Lad

- session 7 turn 11: Hi Priyansh, singing off-key is the best kind! I sing along too, mostly in the kitchen while making tea. It's a joyful kind of dosti, isn't it? Do you write a

Suggested fix: Sharpen Rule 9 with 'NEVER more than ONE question mark per reply'. Optionally a guard that drops trailing questions when count>1.

4. `stale\_phrase\_love\_how\_you\_opener` · severity 3 · freq 18.0% · score 54

Layer: prompt+guard

Maya used the stale phrase tagged 'love_how_you_opener' in 9/50 sessions (18.0%).

Examples:

- session 8 turn 4: I love how you explained it - makes me think of chai too! Just the right balance of sweet and bitter. Your gloves sound like they'll be a hit once ready. What's
- session 12 turn 1: Hi Aarti, I love how you created a clever way to remember morphine! Tell me, have you used this trick for other medicines too?
- session 14 turn 1: Hi Aarti, exams are tough, right? I love how you make your own mnemonics - it shows how hard you work. Did you try the 'prop up your heart' one yet?

Suggested fix: Add to forbidden-phrase list in QWEN_EXTRA_RULES Rule 35b; mirror as deterministic regex guard in pg_apply_output_guard.

5. `stale\_phrase\_hindi\_songs\_persona\_cycle` · severity 4 · freq 12.0% · score 48

Layer: prompt+guard

Maya used the stale phrase tagged 'hindi_songs_persona_cycle' in 6/50 sessions (12.0%).

Examples:

- session 3 turn 4: I listen to light music too when I need to focus - usually old Hindi film songs. They calm my mind.

You mentioned instrumental music helps you think - what kind

- session 20 turn 3: Hi Aarti, I take short walks or sip tea while listening to old Hindi songs. Even 5 minutes of deep breathing helps. What song do you like best for calm?
- session 23 turn 3: Pankaj Udhas is a good choice - very calming! Chai is perfect too, especially with a break. I like to take short walks or listen to old Hindi film songs. What's

Suggested fix: Add to forbidden-phrase list in QWEN_EXTRA_RULES Rule 35b; mirror as deterministic regex guard in pg_apply_output_guard.

6. `stale\_phrase\_echo\_praise` · severity 4 · freq 4.0% · score 16

Layer: prompt+guard

Maya used the stale phrase tagged 'echo_praise' in 2/50 sessions (4.0%).

Examples:

- session 11 turn 3: You said "pharmacology" perfectly! To remember tricky terms, I connect them to daily life. For example, "aspirin" sounds like "aspirants" - people who study hard
- session 18 turn 8: I get that - study time can take over! You said "prop up my heart" perfectly. Speaking of which, what gives you the biggest energy lift during study breaks - music

Suggested fix: Add to forbidden-phrase list in QWEN_EXTRA_RULES Rule 35b; mirror as deterministic regex guard in pg_apply_output_guard.

7. `stale\_phrase\_self\_intro\_in\_reply` · severity 5 · freq 2.0% · score 10

Layer: prompt+guard

Maya used the stale phrase tagged 'self_intro_in_reply' in 1/50 sessions (2.0%).

Examples:

- session 46 turn 2: *Hi Neha, I'm Miss Maya. So glad you're here to practice English! How was your day today?*

Suggested fix: Add to forbidden-phrase list in QWEN_EXTRA_RULES Rule 35b; mirror as deterministic regex guard in pg_apply_output_guard.